

A MACE potential for the Li / LLZO garnet interface

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Six rounds of active learning, each spending a fixed DFT-FE budget on the configurations **farthest in latent space from everything already labelled**. The reference method is the differentiator: real-space DFT-FE lets us label **open, production-scale interfaces** that a plane-wave code cannot represent.

CONFIGURATIONS SCREENED
414,019

across five FPS rounds

DFT-FE LABELLED
108

training set — each its own calculation

LOCAL ENVIRONMENTS
73,634

what the model learns from

HELD OUT
42

35 + 7, never trained on

MEAN FORCE ERROR
120.5 → 72.0

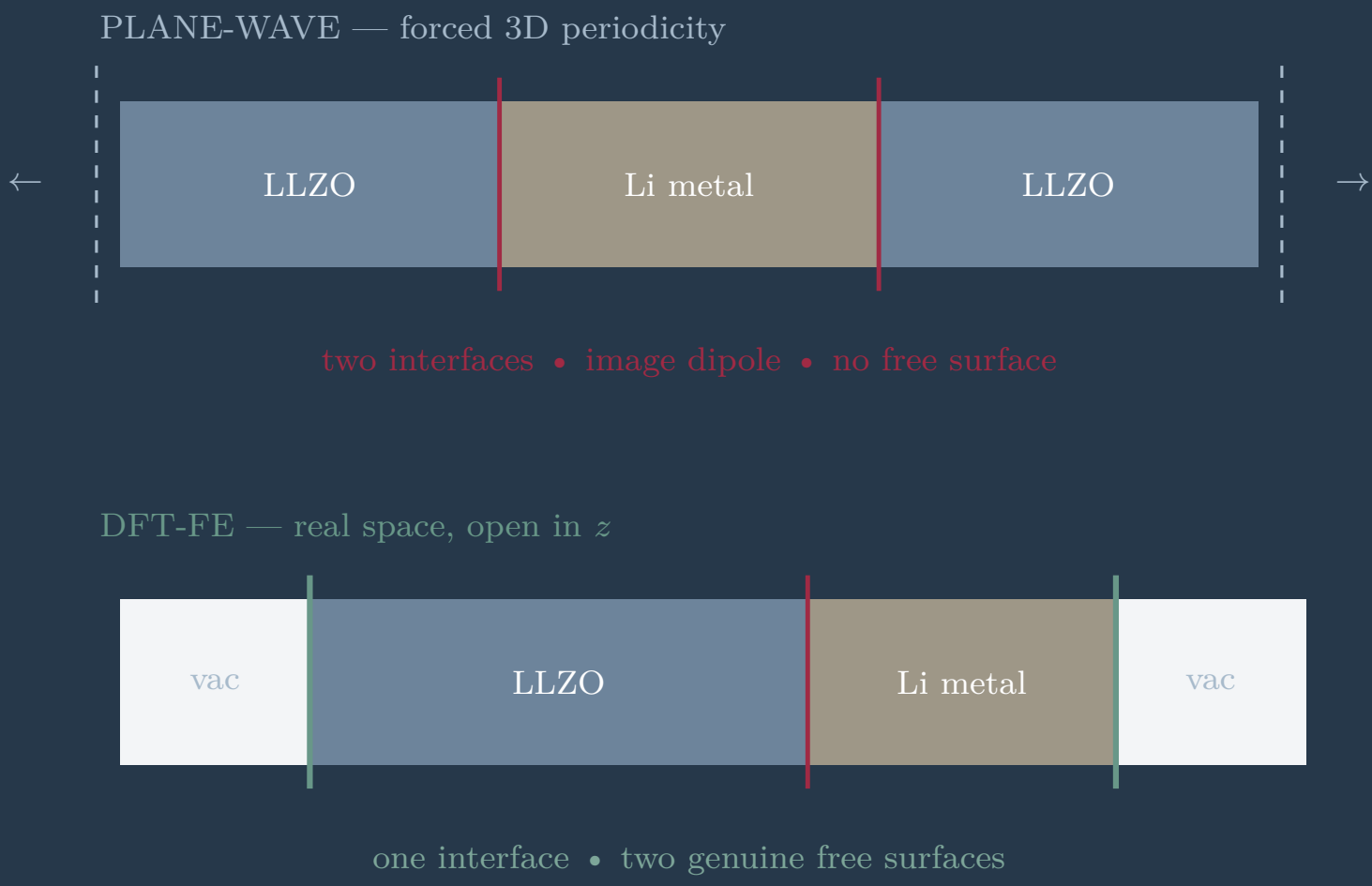
meV/Å • universal → it6

00 – WHY DFT-FE

Nearly every MLIP for solid-state interfaces is trained on what plane-wave DFT can afford: **small, fully-periodic cells** — a constraint that decides what the dataset can contain, and what the potential can honestly be used for. A real-space solver lifts it on three fronts at once.

01 • BOUNDARY CONDITIONS

One real interface, two free surfaces



Every structure is `pbc = [True, True, False]`. 3D periodicity would force a **sandwich** — a second interface, an image dipole, a constrained slab thickness. DFT-FE removes all three, so **the thing modelled is the interface that actually exists**.

02 • LABEL QUALITY

The labels are not the limiting factor

0.06

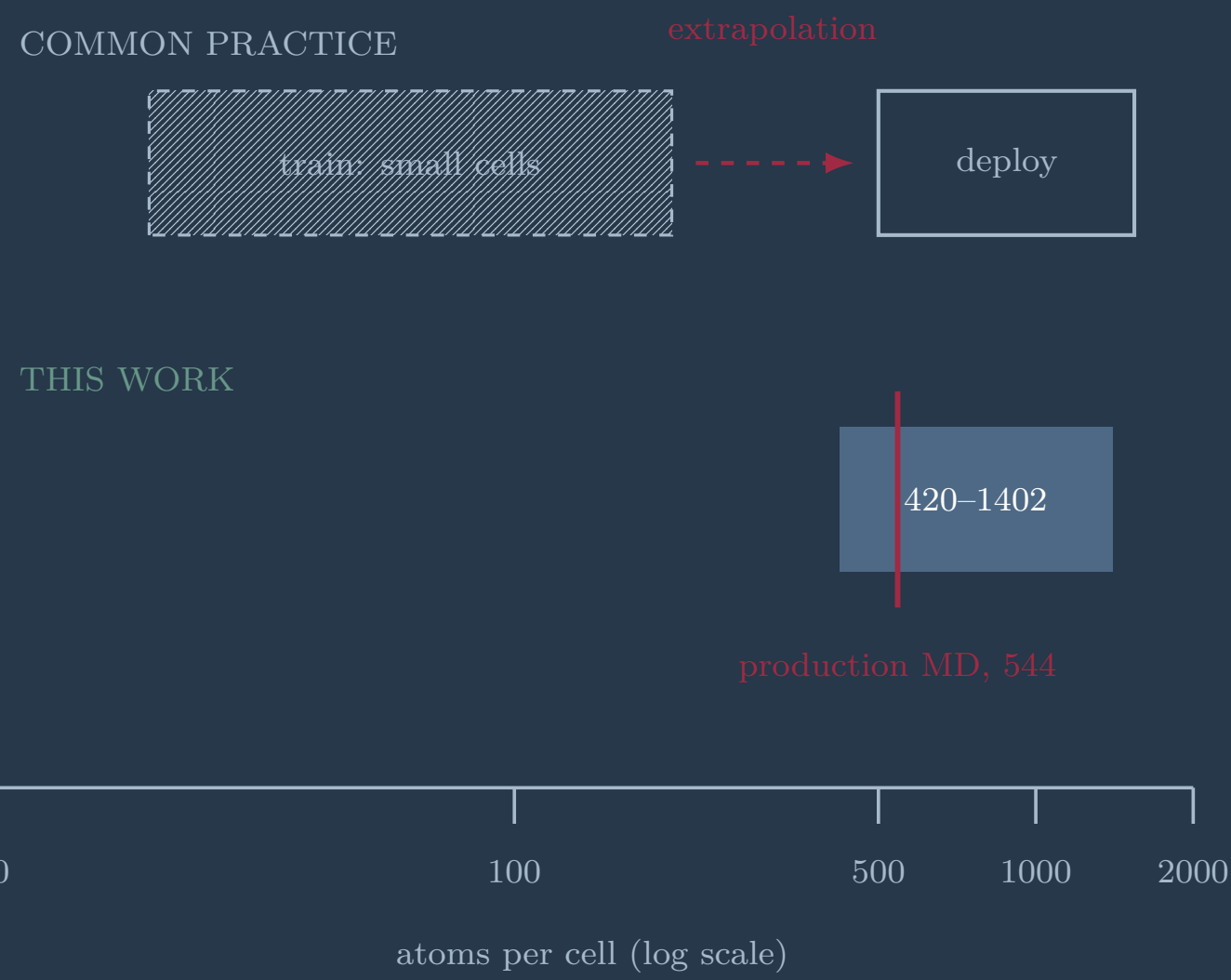
meV/atom • 0.12 meV/Å on forces

A duplicated structure, recomputed on a larger real-space domain, labelled the same physics twice. That spread is the **noise floor of the reference method** — measured, not assumed — and sits **≈50× below** current model error: accuracy is model-limited, not label-limited.

The transfer is real work — trained on periodic bulk, the foundation model leaves the vacuum-facing LLZO layer at **≈2× the frame-mean force error**.

03 • SCALE

Trained where it is deployed

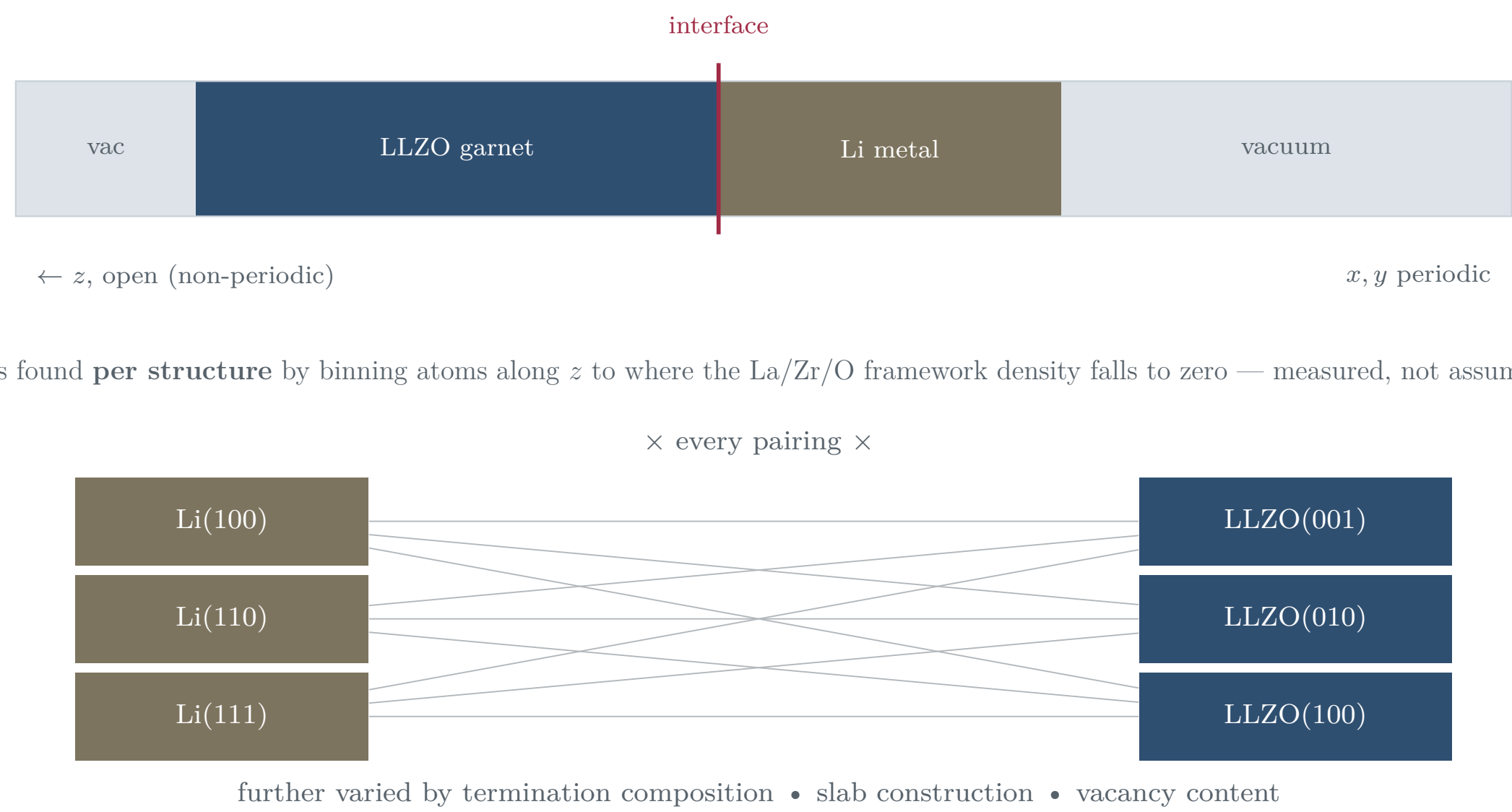


MACE sees each atom inside its cutoff sphere, so **every atom in a labelled cell is a training example**: 108 structures buy **73,634** local environments, in cells of 420–1,402 atoms.

The usual route trains on small cells, then runs production MD at 500–1,500 atoms — a **scale extrapolation**. We train and deploy on the same scale. (Our band measured; comparison band schematic.)

01 – THE SYSTEM

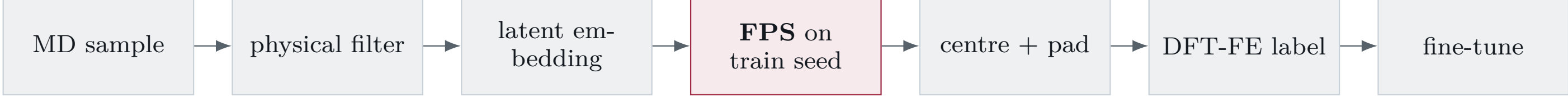
An open slab, periodic only in plane



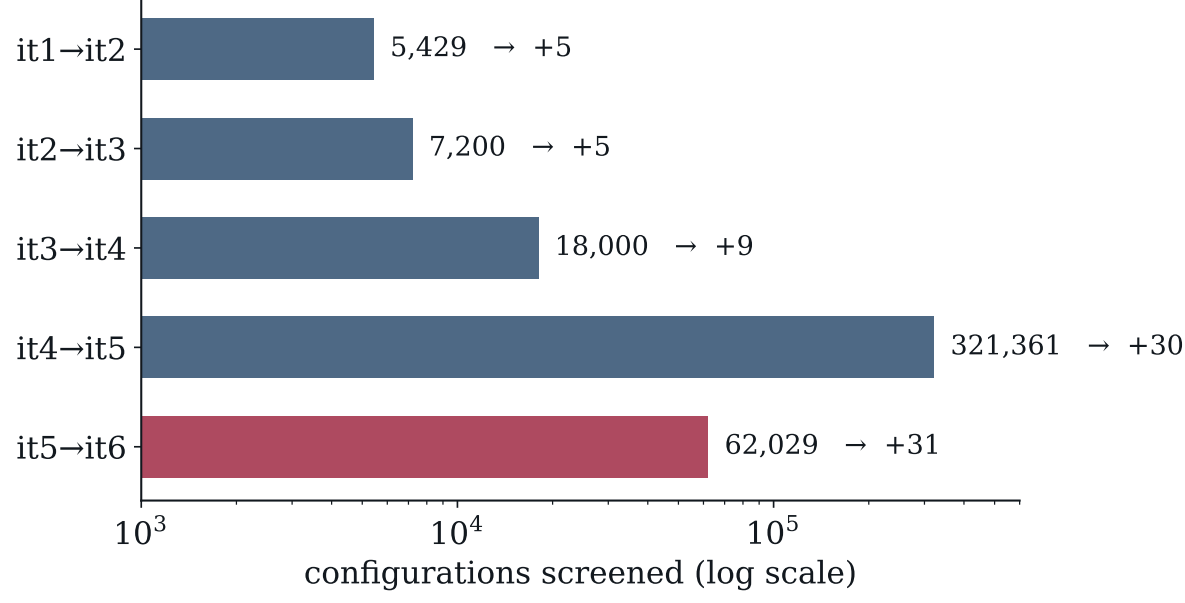
Every number here is averaged over this full multi-orientation pool.

02 – THE METHOD

One round, repeated it0 → it6



Seeding FPS with the *existing* training set means a near-duplicate can never be picked — de-duplication falls out of the method. Selection stops when the worst-covered candidate falls inside the **d99** radius: ρ drops 0.0510 → 0.0207 past d99 = 0.0208, taking pool coverage **98.999% → 100%**.

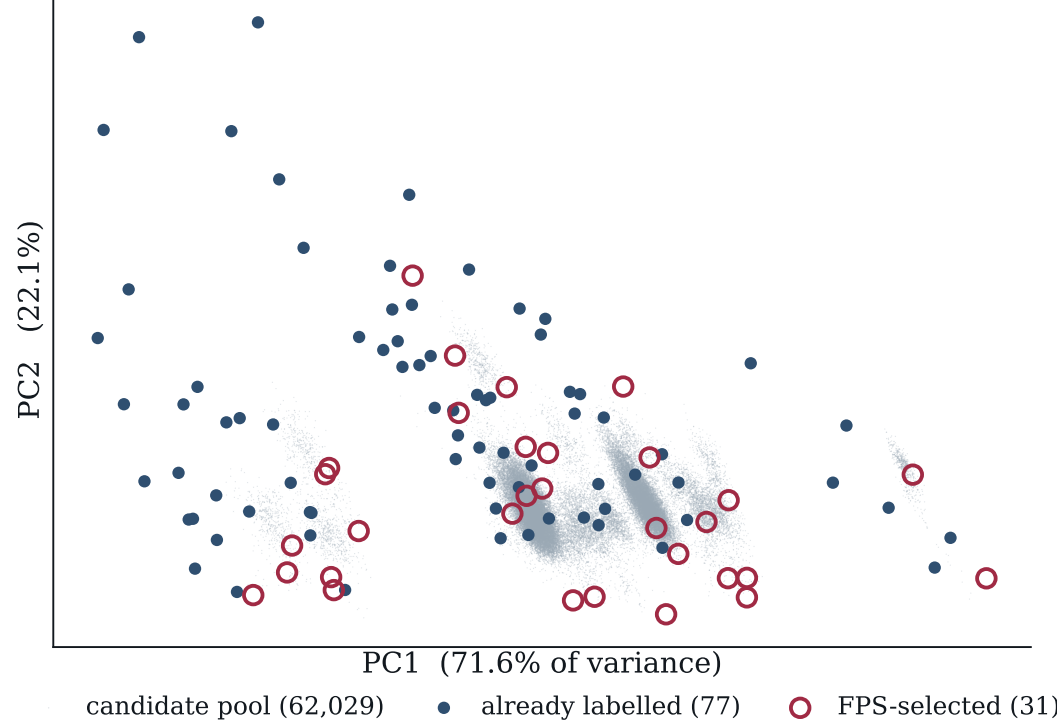


Pool size swings two orders of magnitude — the method asks how much of the pool is already covered, not what the budget is. **All 108 carry their own DFT-FE calculation**.

03 – SELECTION, ONE ROUND IN FULL

Where the 31 picks actually land

PC1 × PC2 of L2-normalised 256-d MACE latents • 93.7% of variance • real it5→it6 round

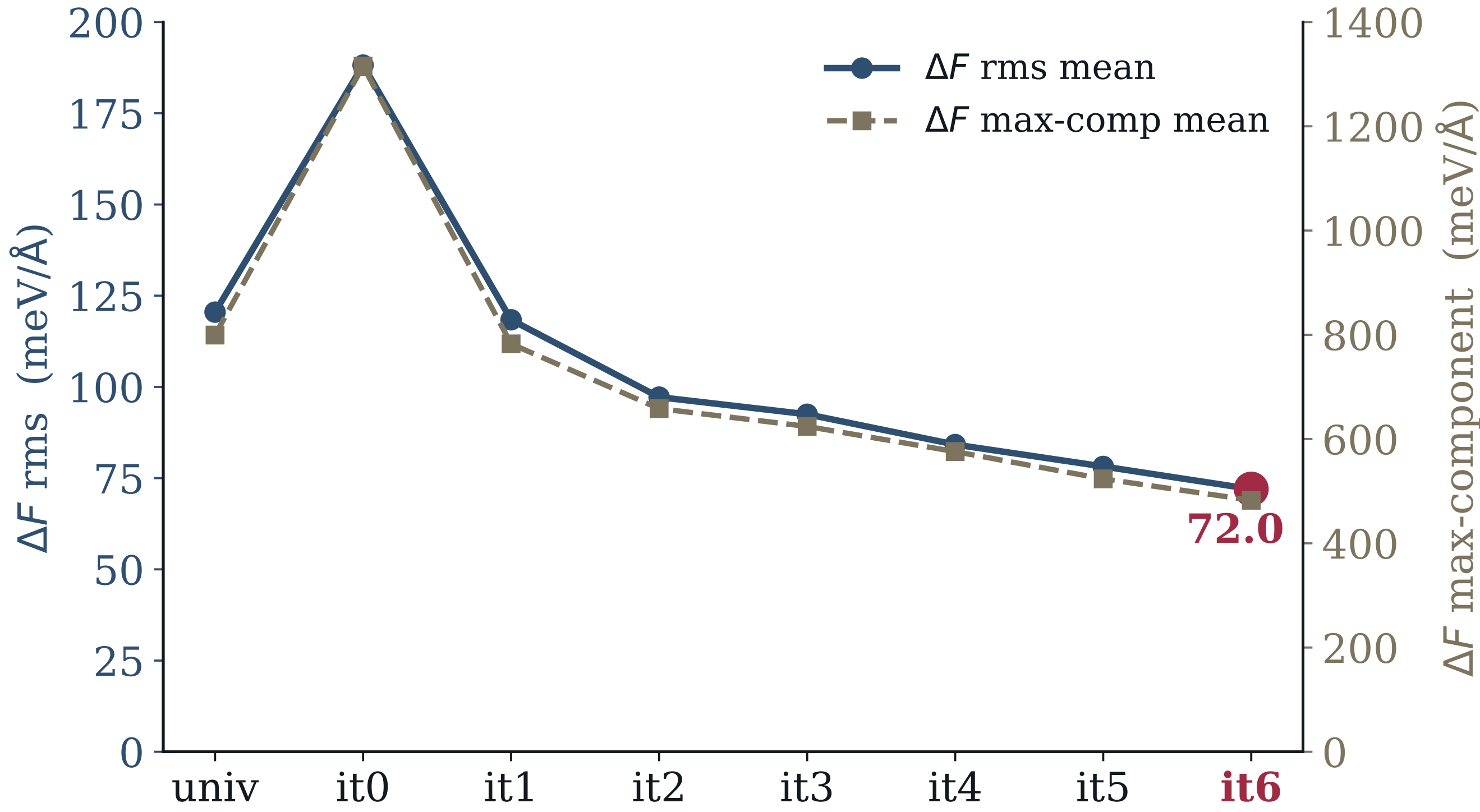


Every point is a real MD configuration. The seed sits in the dense core; the picks sit on the sparse rim. **Coverage-driven, not error-driven** — the criterion never looks at the model's error. $\delta_k = \min_{a \in S} \|Z_k - Z_a\|$, $\rho = \max_k \delta_k$.

04 – ACCURACY ACROSS ITERATIONS

Error falls every round

34-structure held-out set • meV/Å • lower is better



Two metrics: ΔF rms dilutes a bad component across 500+ atoms × 3, while ΔF max-comp is the worst single component anywhere — what governs MD and NEB stability.

05 – FULL COMPARISON

34-structure held-out set • ΔE meV/atom • ΔF meV/Å

model	ΔE		ΔF rms		ΔF max-comp	
	mean	max	mean	max	mean	max
universal [†]	15.8	30.2	120.5	201.9	799.5	1665.9
it0	9.8	60.5	188.2	330.4	1315.5	2472.6
it1	5.5	20.3	118.4	167.8	782.6	1284.8
it2	3.8	19.3	97.2	182.7	658.3	1250.6
it3	5.1	25.9	92.5	158.5	624.2	1111.7
it4	5.0	39.2	84.2	128.9	575.9	914.5
it5	4.7	35.8	78.2	112.9	523.6	900.1
it6	4.5	19.0	72.0	99.9	482.5	746.8

it6 wins every column. [†]universal's ΔE is a cohesive-energy difference — the two use different absolute-energy conventions.

06 – SCOPE & INDEPENDENT CHECK

Domain of validity

paired it5→it6	ΔF rms	p
it5, in-domain 33	77.2	
it6, in-domain 33	72.4	0.008
energy, same set	flat	0.50

Training covers 0.5 and 1 vacancy/f.u. The 2 vac/f.u. tier was cut at pool construction — half blew up above 3000 K — so **zero were ever trained on**.

Separate trajectory, no time drift

model	ΔE	ΔF rms	ΔF me
universal	16.1	136.6	908.7
it4	1.2	76.0	538.7
it5	1.2	83.9	564.9
it6	1.6	80.0	519.2

7 DFT-FE snapshots from a 1 ns / 1100 K run, never trained on. Error at 1000 ps (87.7) is no worse than at 0 ps (49.3).

07 – STATUS & NEXT

- Held-out accuracy, best of all iterations
- Trajectory set, best or tied, no time drift
- Label noise measured, $\approx 50\times$ below model error
- z -padding convergence study
- Transport validation, in progress
- Multi-temperature MD for a measured activation energy